

Prof. T. V. K. Singh

Prof. T. V. K. Singh graduated with distinction in B.Sc. (Agriculture) from the College of Agriculture, Rajendranagar, APAU (1971–1976). He completed his M.Sc. in Entomology (1976–1978) with a State Fellowship from Andhra Pradesh Agricultural University, and later earned his Ph.D. (1986–1989) from the Indian Agricultural Research Institute (IARI), New Delhi, under the ICAR Senior Research Fellowship. He also received specialized training in Insecticide Resistance from the University of Florida, USA.

Over the course of an illustrious career, Prof. Singh has served in several key academic and administrative positions at ANGRAU and PJTSAU — including Senior Professor, Vice Principal, Principal, Associate Director of Research, Dean of Agriculture, Director of Extension, and Comptroller.

He is a Fellow of the Royal Entomological Society (London), the Linnean Society (London), the Entomological Society of India, the Plant Protection Association of India, and the Indian Society of Oilseed Research. His distinguished teaching career was recognized with the **University Meritorious Teacher Award (1996)** from the Governor of Andhra Pradesh and the **State Meritorious Teacher Award (2007)** from the Government of Andhra Pradesh, presented by (Late) Dr. Y. S. Rajasekhara Reddy, the then Chief Minister.

A veteran academician and researcher, Prof. Singh has more than three decades of experience in higher education, specializing in insect ecology, toxicology, population dynamics, economic entomology, and eco-friendly pest management strategies. He has played a pivotal role in designing, teaching, and modernizing Entomology curricula for Diploma, Undergraduate, Master's, and Doctoral programs, and has developed several practical manuals in alignment with ICAR guidelines.

His extensive research portfolio encompasses pest management across legumes, fruits, and vegetables. His pioneering contributions to insect ecology and insecticide resistance management are evidenced by high-impact publications and books with reputed publishers. Prof. Singh's leadership and innovation have made him an invaluable asset to institutions such as ANGRAU, PJTSAU, and ICRISAT.

He has mentored over **15 M.Sc.** and **5 Ph.D.** scholars as Major Advisor, and served as Co-Chair for more than **30 M.Sc.** and **10 Ph.D.** dissertations. His students' collective work has significantly advanced knowledge in **Insect Ecology, IPM, Economic Entomology, and Insecticide Resistance**.

Prof. Singh's seminal research includes the study of distribution patterns of safflower aphid, diamondback moth, cabbage leaf webber, brinjal shoot and fruit borer, and sunflower thrips—forming the basis for pest sampling plans. His life and fertility table analyses for major pests such as *Plutella xylostella*, *Crociodolomia binotalis*, *Aproaerema modicella*, and *Spodoptera litura* on Bt and non-Bt cotton identified key mortality factors and yield–infestation dynamics.

He developed sequential sampling plans for groundnut leafhopper, groundnut thrips, and safflower aphid, enabling optimal timing of insecticide applications. His stochastic models for predicting *Helicoverpa armigera* infestations in Bt cotton were successfully validated under field conditions, alongside migration studies of *Spodoptera litura*.

A landmark achievement was establishing an Insecticide Resistance Monitoring Laboratory at the university, which operated for over 15 years. From this, the **F₂ screen method** for estimating resistance to Bt cotton in India was developed. His team estimated *Cry1Ac* resistance alleles and associated fitness costs in *H. armigera* populations from Telangana and Andhra Pradesh, and explored molecular characterization of resistance to *Cry2Ab* using RAPD primers. His applied research translated into effective IPM packages most notably for red hairy caterpillar and *S. litura* on castor leading to significant pest suppression in farmers' fields.

Prof. Singh has served as Principal Investigator for several national and international research projects funded by ICAR–CICR, BTU–IPE, Osmania University, NTGCF–Anand, and DST–Government of India, covering diverse themes such as insecticide resistance, pheromones, botanicals, and vermiculture technology.

In his administrative roles, he chaired the Educational Technology Cell, contributed as a member of the State Crisis/Disaster Management Team, and served on the IV and V Deans' Committees for finalization of undergraduate curricula. He has also served on numerous national monitoring and review committees, including QRTs for Rice, Cotton, and IPM, and on confidential assignments for ICAR, ASRB, and several agricultural universities.

Prof. Singh's dedication to teaching, research, and extension has earned him wide recognition. He has authored **over 170 research papers, 13 books, 4 book chapters**, edited **2 books**, and delivered **25 keynote/lead papers** at national and international platforms. He has participated in **35 symposia, seminars, and workshops**, and organized **five international seminars and three national conferences** including the acclaimed *Entomology 2018: Advances and Challenges*, *Entomology 2020: Beyond COVID-2019*, and *Entomology 2022: Innovation & Entrepreneurship*.

Prof. Singh was Consultant for writing report on "***Study on impact of indiscriminate use of chemical fertilizers and pesticides***" was conducted during *kharif* 2018-19 in 7 states covering seven State Agricultural Universities (SAUs) and one State Horticultural University (SHU) Funded by Department of Agriculture & Cooperation and Farmers Welfare, Ministry of Agriculture, Govt. of India , carried out by **NIPHM as Nodal Agency**

During February 2026 , he played an important role in organising National Conference on "**Innovations in Plant Health Management for Resilient Agriculture and Agri-Export Promotion (IPHMRA-AEP)**", held during 26th to 28th Feb 2026 at **NIPHM**, Hyderabad.

He was the **first ICAR Emeritus Scientist of PJTSAU**, and the inaugural recipient of the **Dr. Y. Ramachandra Rao Memorial Award** for outstanding ecological and ethological research. In 2006, he had the honor of meeting **Mr. George W. Bush**, President of the United States, to brief him on the state's plant protection initiatives.

As the university's coordinator for the **INTAG Program of Cornell University (USA)**, Prof. Singh played a key role in advancing international collaboration. He also organized numerous *Kisan Melas* and trained farmers in IPM practices for field crops, fruits, and vegetables.

Post-retirement, as a **Consultant at ICRISAT (2018–2020)** under the **SERP initiative (Department of Rural Development, Govt. of Andhra Pradesh)**, he conducted **95 training programs** for over **165 Farmer Producer Organizations** across all districts of Andhra Pradesh benefiting nearly **4,000 farmers** on crop production and pest management in major crops including paddy, maize, groundnut, cotton, chickpea, red gram, chilli, mango, turmeric,

and coconut. In 2021, he also trained farmers and officials in Odisha's Sundargarh district on improved paddy and groundnut technologies.

Prof. Singh is a **Life Member** of over a dozen professional societies, including the Society of Pesticide Science, Plant Protection Association of India, Entomological Society of India, Indian Science Congress Association, Society for Biocontrol Advancement, and Society of Biopesticide Sciences.

He continues to contribute actively as **Chief Editor of the Indian Journal of Plant Protection, Editor of Indian Journal of Ecology**, Editorial **Board Member** of several leading journals (including *Current Science*, *Indian Journal of Entomology*, *Shashpa*, and *Indian Journal of Plant Protection*), and **Chief Content Reviewer** for the *Plantix* mobile application developed by PEAT GmbH, Berlin.

Renowned for his scholarly rigor, literary passion, and enduring mentorship, Prof. Singh remains deeply engaged in advancing agricultural entomology through lectures, research guidance, and farmer outreach.